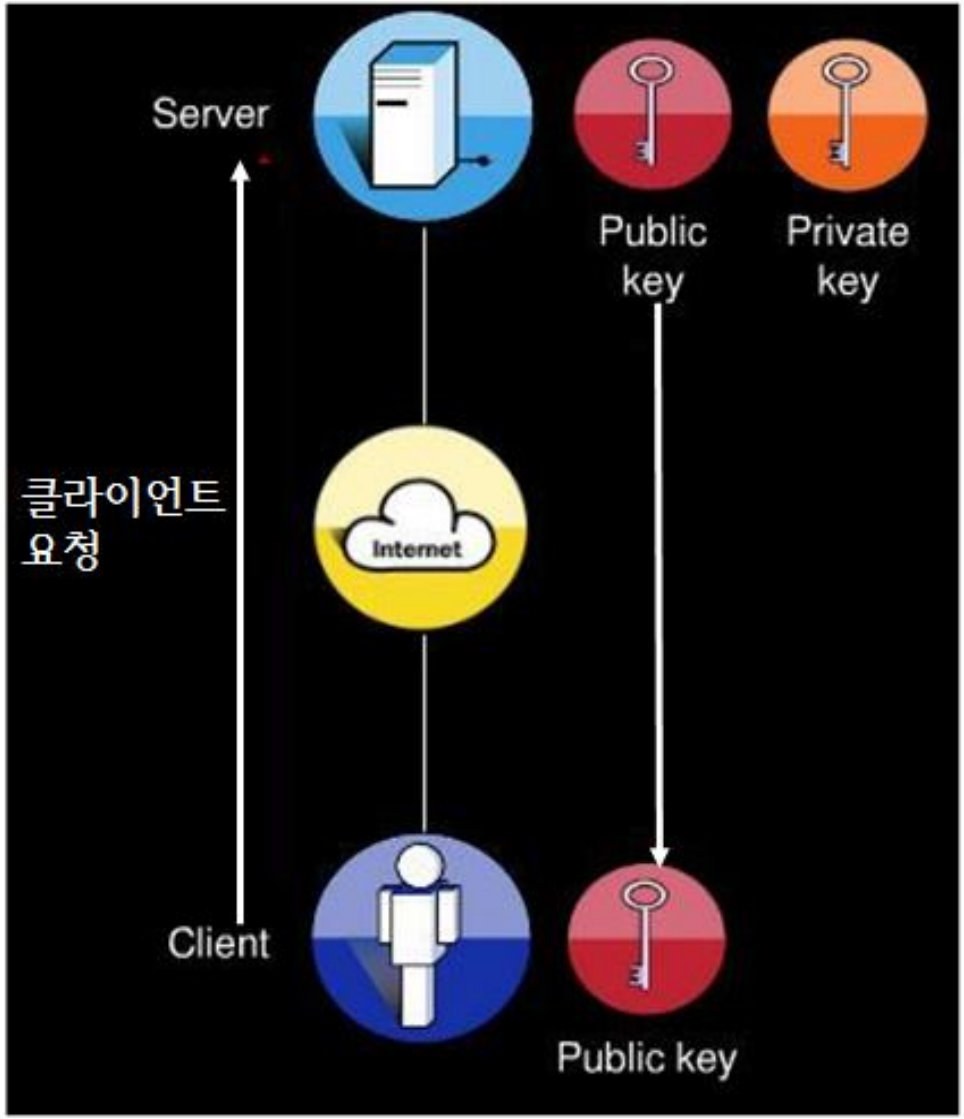
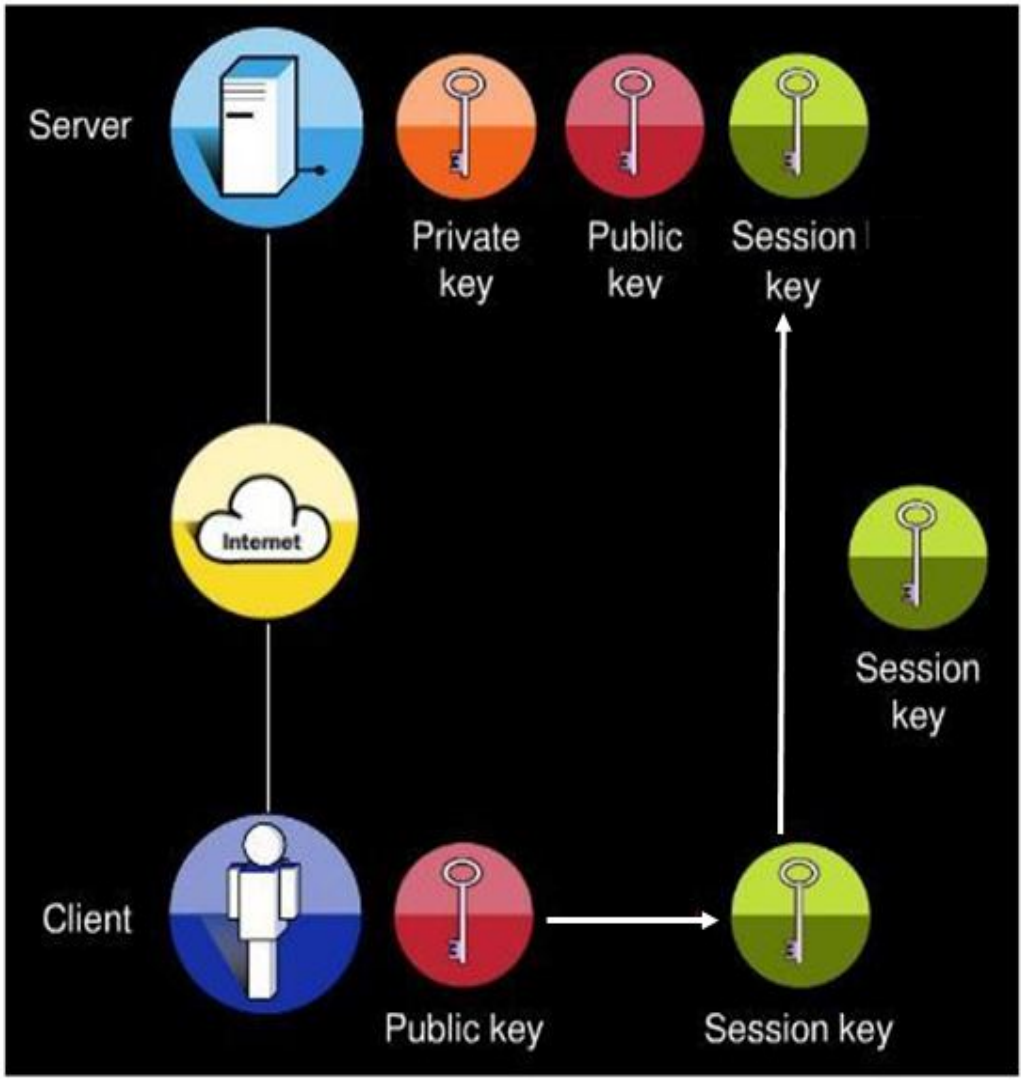
SSL/TLS Protocol

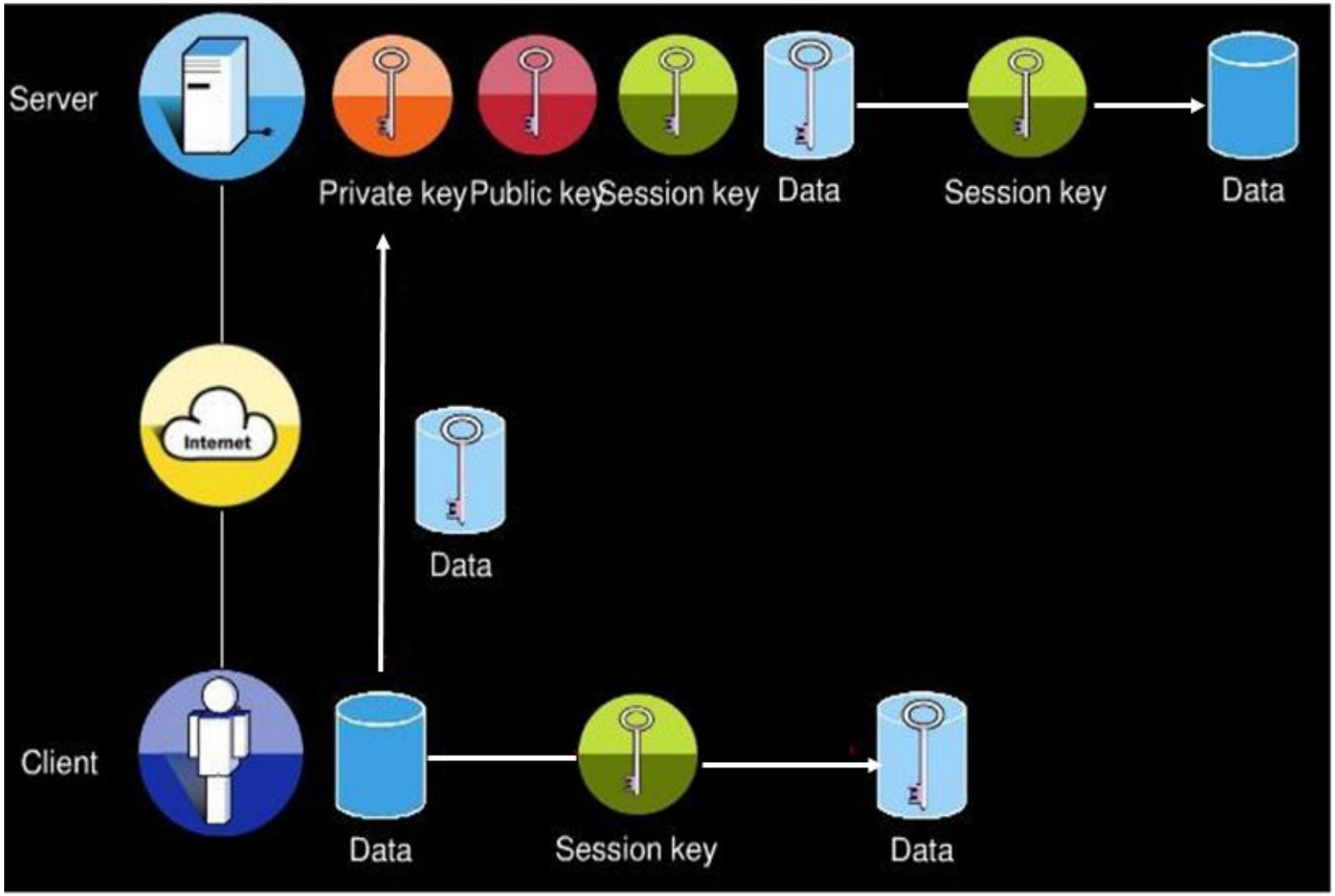
SSL/TLS Processing



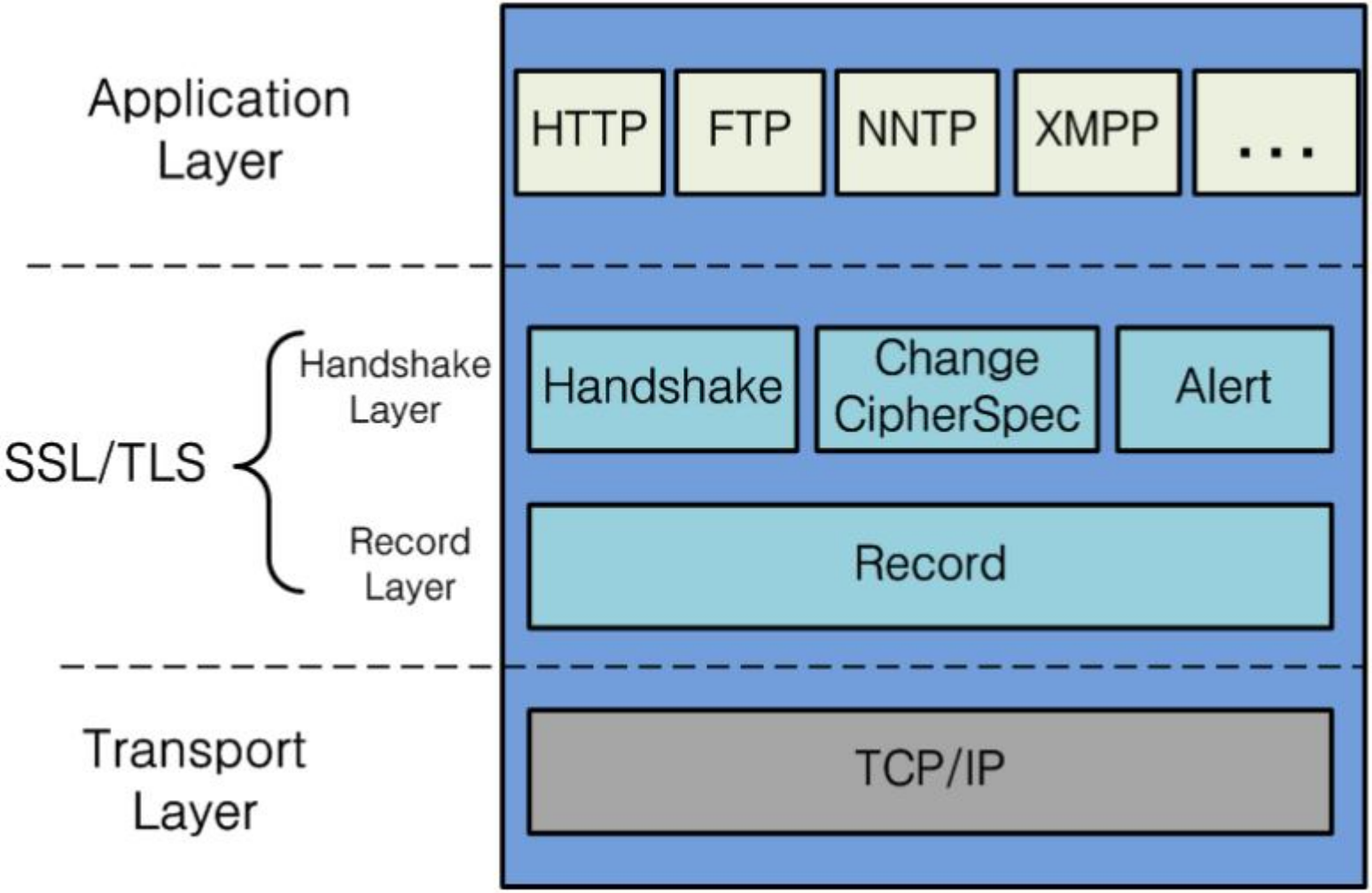
SSL/TLS Processing



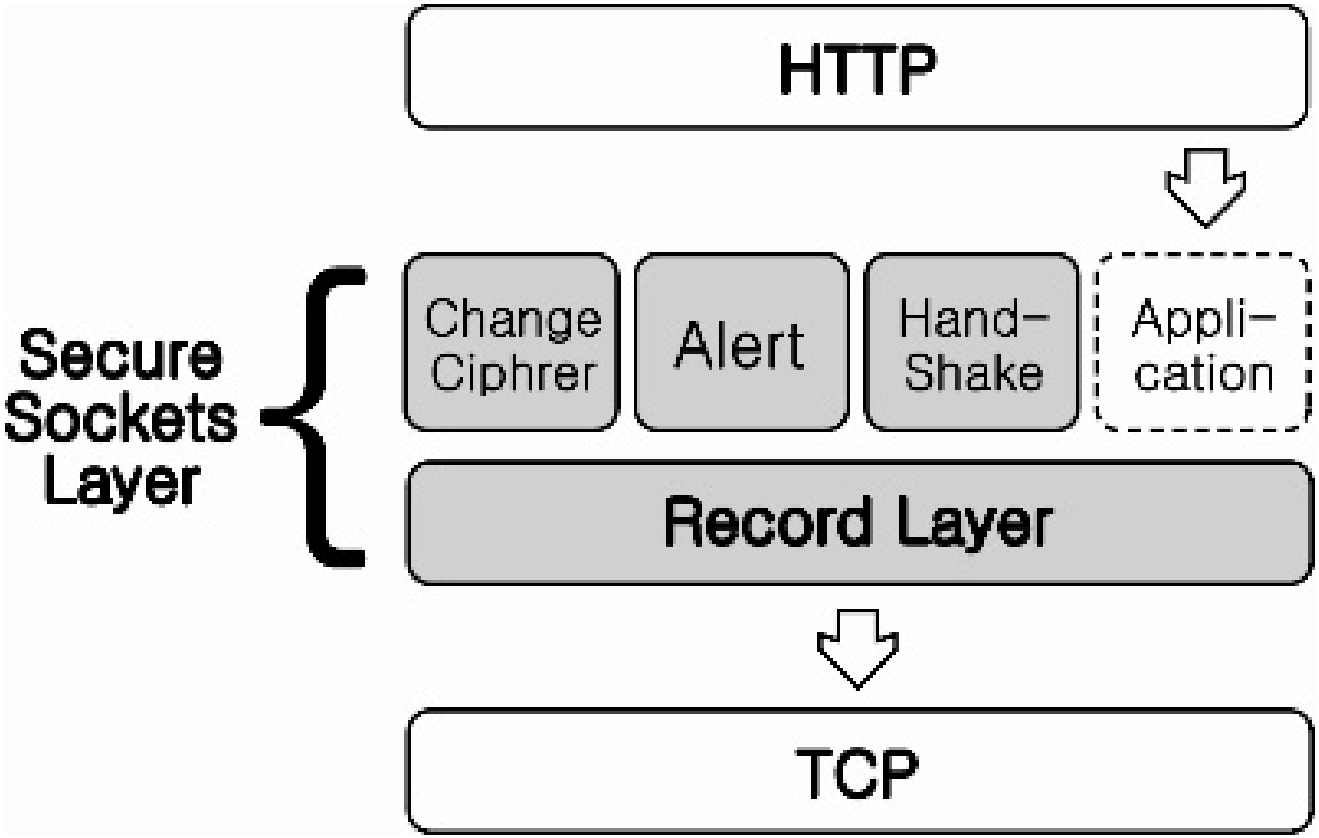
SSL/TLS Processing



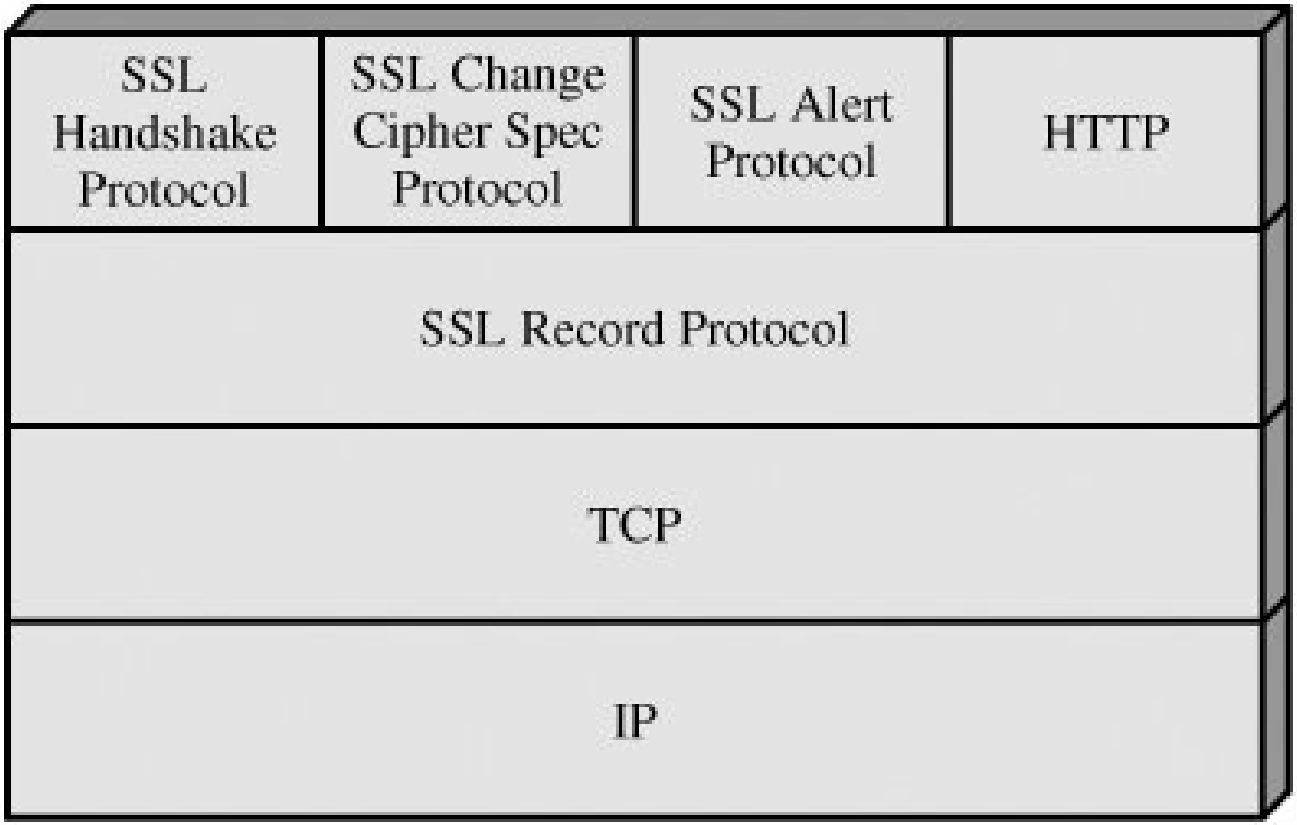
SSL/TLS Protocol stack



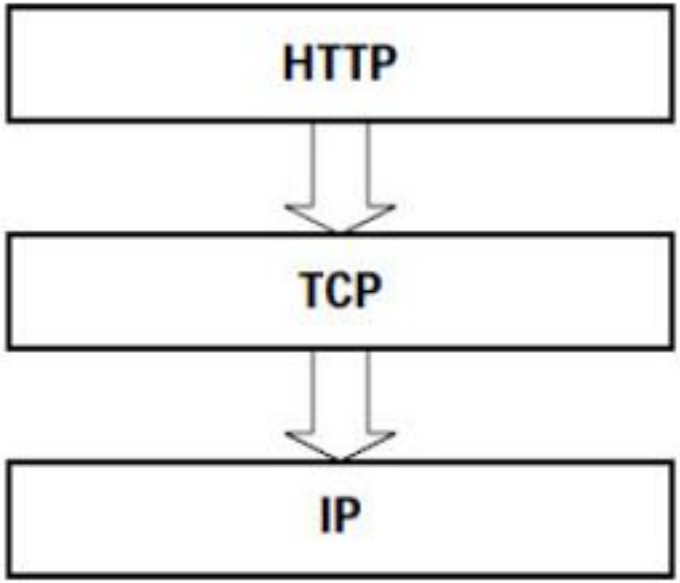
SSL/TLS Protocol stack



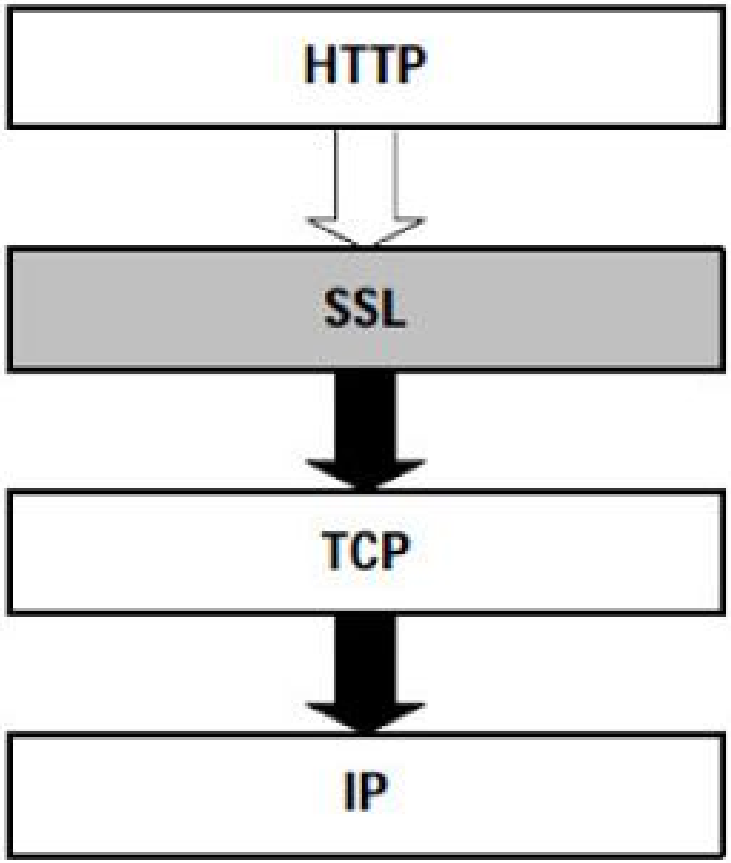
SSL/TLS Protocol stack



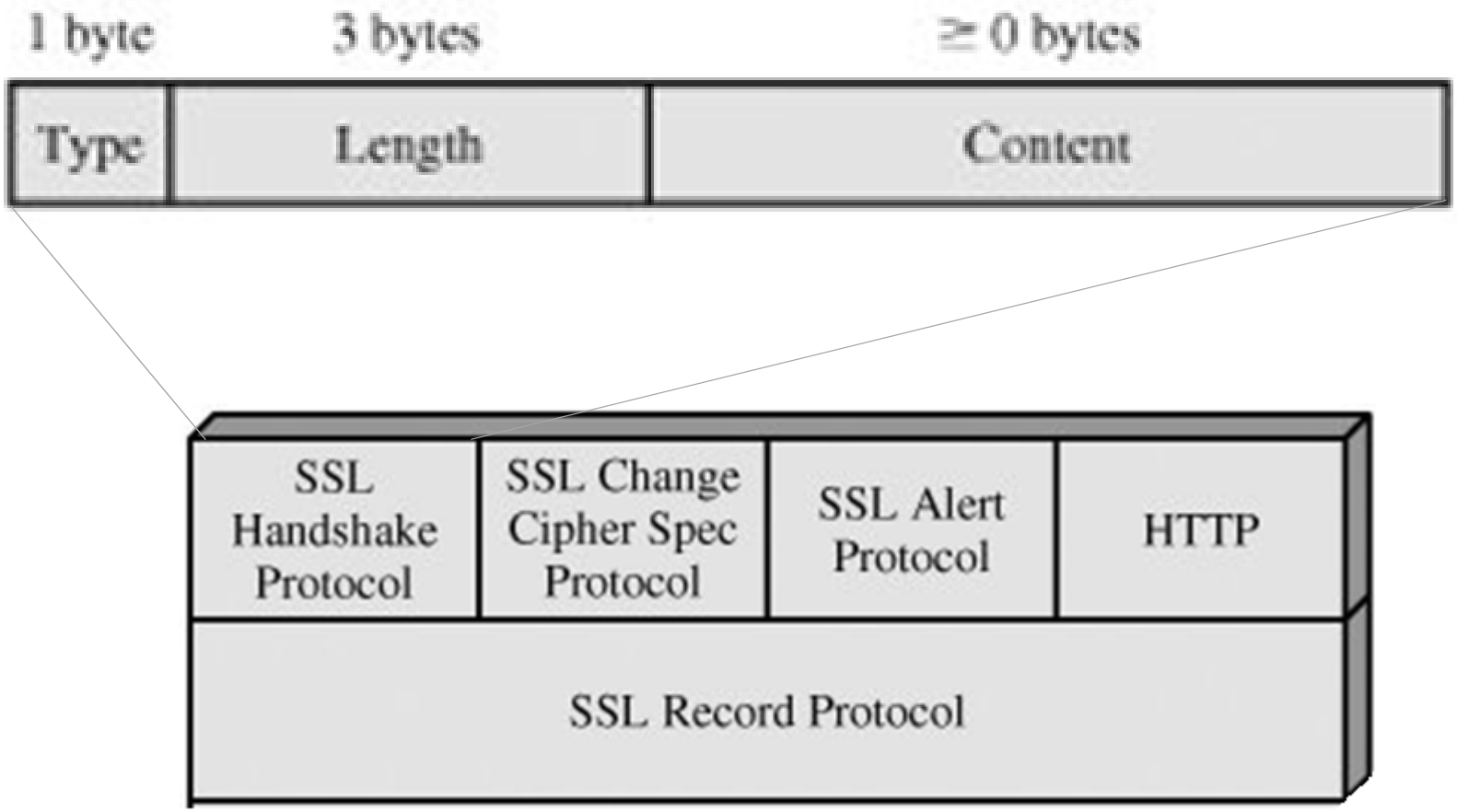
HTTP



HTTPS



1) Handshake protocol



facebook_login.pcap

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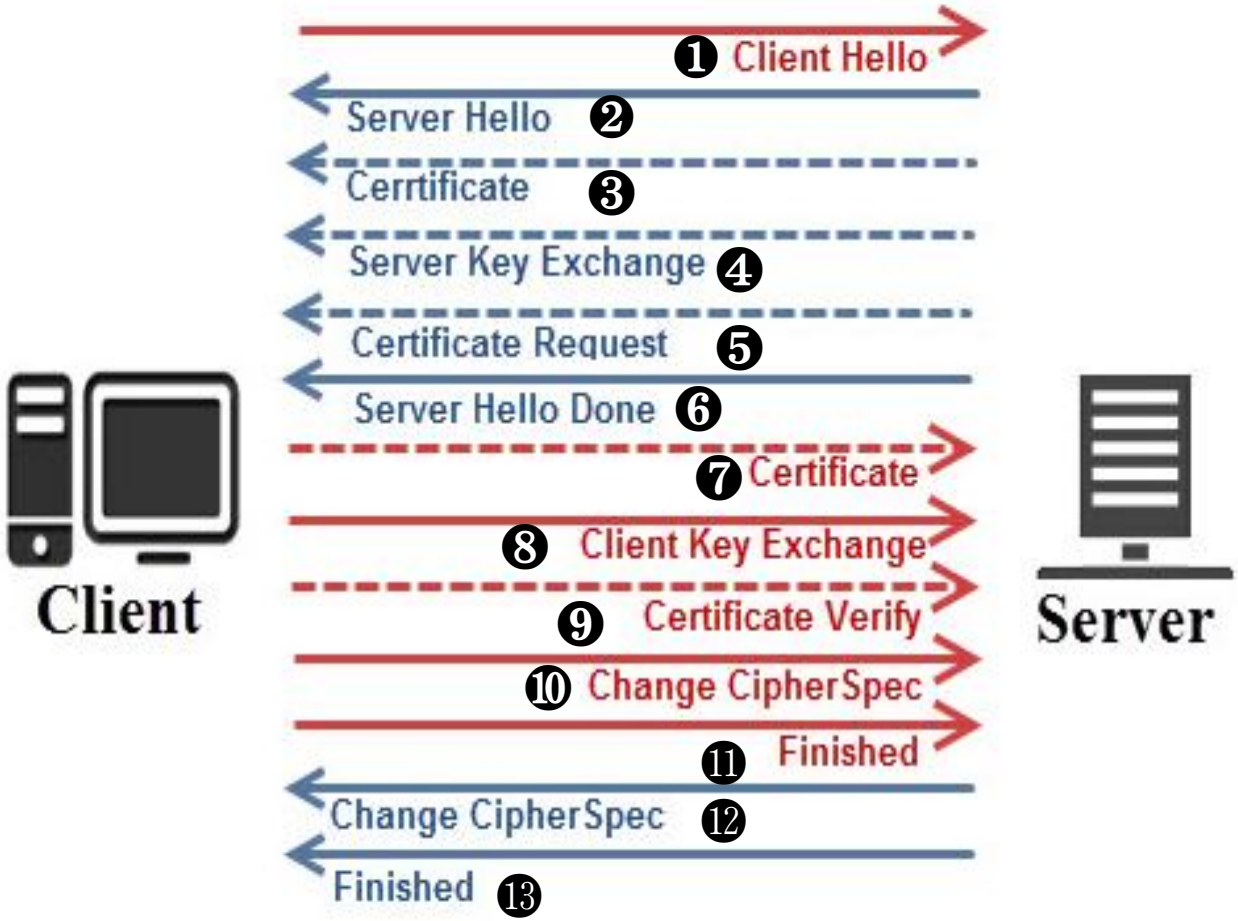
No.	Time	Source	Destination	Protocol	Total Length	Info
1	0.000000s	172.16.0.122	69.63.180.173	TCP	60	54595 → 443 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3019897…
2	0.089900s	69.63.180.173	172.16.0.122	TCP	64	443 → 54595 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 TSval=347…
3	0.000033s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=301989735 TSecr=347912…
4	0.000343s	172.16.0.122	69.63.180.173	TLSv1	221	Client Hello
5	0.089522s	69.63.180.173	172.16.0.122	TLSv1	989	Server Hello, Certificate, Server Hello Done
6	0.000031s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=170 Ack=938 Win=7744 Len=0 TSval=301989758 TSecr=34…
7	0.002848s	172.16.0.122	69.63.180.173	TLSv1	234	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
8	0.090444s	69.63.180.173	172.16.0.122	TLSv1	95	Change Cipher Spec, Encrypted Handshake Message
9	0.000533s	172.16.0.122	69.63.180.173	TLSv1	1034	Application Data
10	0.189619s	69.63.180.173	172.16.0.122	TCP	52	443 → 54595 [ACK] Seq=981 Ack=1334 Win=5473 Len=0 TSval=3479126142 TSecr=…
11	0.073201s	69.63.180.173	172.16.0.122	TLSv1	1233	Application Data
12	0.011497s	172.16.0.122	69.63.190.22	HTTP	679	GET /home.php? HTTP/1.1



ssl

No.	Time	Source	Destination	Protocol	Total Length	Info
4	0.000000s	172.16.0.122	69.63.180.173	TLSv1	221	Client Hello
5	0.089522s	69.63.180.173	172.16.0.122	TLSv1	989	Server Hello, Certificate, Server Hello Done
7	0.002879s	172.16.0.122	69.63.180.173	TLSv1	234	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
8	0.090444s	69.63.180.173	172.16.0.122	TLSv1	95	Change Cipher Spec, Encrypted Handshake Message
9	0.000533s	172.16.0.122	69.63.180.173	TLSv1	1034	Application Data
11	0.262820s	69.63.180.173	172.16.0.122	TLSv1	1233	Application Data
63	5m 1.502778s	172.16.0.122	69.63.180.173	TLSv1	75	Encrypted Alert

1 단계. Client Hello message

```
▷ Frame 4: 235 bytes on wire (1880 bits), 235 bytes captured (1880 bits)
▷ Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: Cisco_31:07:33 (00:26:0b:31:07:33)
▷ Internet Protocol Version 4, Src: 172.16.0.122, Dst: 69.63.180.173
▷ Transmission Control Protocol, Src Port: 54595, Dst Port: 443, Seq: 1, Ack: 1, Len: 169
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Handshake Protocol: Client Hello
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 164
    ◢ Handshake Protocol: Client Hello
        Handshake Type: Client Hello (1)
        Length: 160
        Version: TLS 1.0 (0x0301)
      ▷ Random: 4bba350339dc8387b20a0c5cfa490f4807d25f05c6c4cbdc...
        Session ID Length: 0
        Cipher Suites Length: 70
      ▷ Cipher Suites (35 suites)
        Compression Methods Length: 1
      ▷ Compression Methods (1 method)
        Extensions Length: 49
      ▷ Extension: server_name (len=23)
      ▷ Extension: supported_groups (len=8)
      ▷ Extension: ec_point_formats (len=2)
      ▷ Extension: SessionTicket TLS (len=0)
```

content

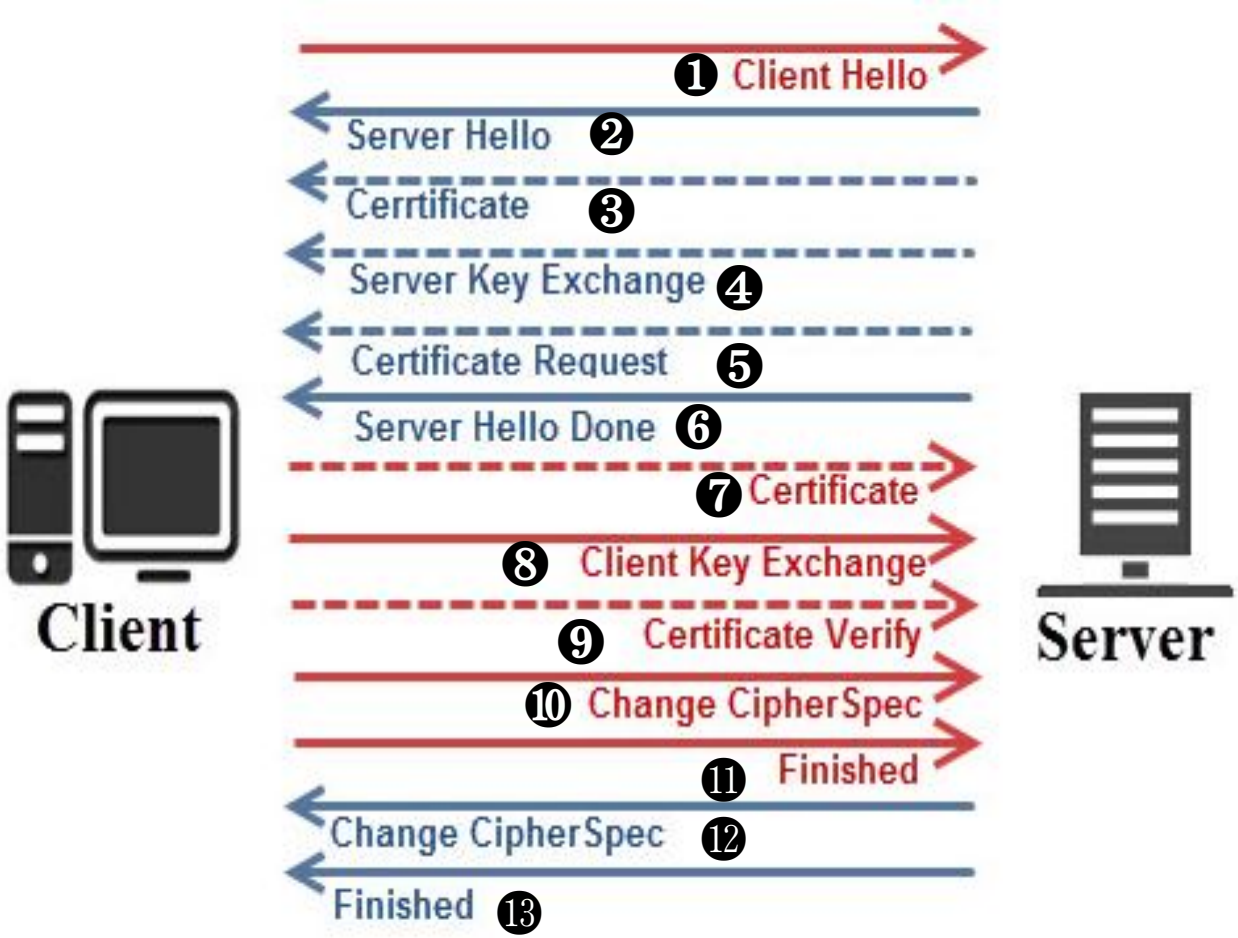
Protocol Suite

```
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Handshake Protocol: Client Hello
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 164
    ◢ Handshake Protocol: Client Hello
        Handshake Type: Client Hello (1)
        Length: 160
        Version: TLS 1.0 (0x0301)
      ◢ Random: 4bba350339dc8387b20a0c5cfa490f4807d25f05c6c4cbdc...
          GMT Unix Time: Apr  6, 2010 04:07:47.000000000 대한민국 표준시
          Random Bytes: 39dc8387b20a0c5cfa490f4807d25f05c6c4cbdc71fa59e8...
        Session ID Length: 0
        Cipher Suites Length: 70
      ◢ Cipher Suites (35 suites)
          Cipher Suite: TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA (0xc00a)
          Cipher Suite: TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA (0xc014)
          Cipher Suite: TLS_DHE_RSA_WITH_CAMELLIA_256_CBC_SHA (0x0088)
          Cipher Suite: TLS_DHE_DSS_WITH_CAMELLIA_256_CBC_SHA (0x0087)
          Cipher Suite: TLS_DHE_RSA_WITH_AES_256_CBC_SHA (0x0039)
          Cipher Suite: TLS_DHE_DSS_WITH_AES_256_CBC_SHA (0x0038)
          Cipher Suite: TLS_ECDH_RSA_WITH_AES_256_CBC_SHA (0xc00f)
          Cipher Suite: TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA (0xc005)
          Cipher Suite: TLS_RSA_WITH_CAMELLIA_256_CBC_SHA (0x0084)
          Cipher Suite: TLS_RSA_WITH_AES_256_CBC_SHA (0x0035)
          Cipher Suite: TLS_ECDHE_ECDSA_WITH_RC4_128_SHA (0xc007)
```

```
    ▷ Random: 4bba350339dc8387b20a0c5cfa490f4807d25f05c6c4cbdc...
      Session ID Length: 0
      Cipher Suites Length: 70
    ▷ Cipher Suites (35 suites)
      Compression Methods Length: 1
    ◢ Compression Methods (1 method)
        Compression Method: null (0)
      Extensions Length: 49
    ◢ Extension: server_name (len=23)
        Type: server_name (0)
        Length: 23
      ▷ Server Name Indication extension
    ◢ Extension: supported_groups (len=8)
        Type: supported_groups (10)
        Length: 8
        Supported Groups List Length: 6
      ◢ Supported Groups (3 groups)
          Supported Group: secp256r1 (0x0017)
          Supported Group: secp384r1 (0x0018)
          Supported Group: secp521r1 (0x0019)
    ◢ Extension: ec_point_formats (len=2)
        Type: ec_point_formats (11)
        Length: 2
        EC point formats Length: 1
      ▷ Elliptic curves point formats (1)
    ◢ Extension: SessionTicket TLS (len=0)
        Type: SessionTicket TLS (35)
        Length: 0
        Data (0 bytes)
```

1 단계. Server Hello message

```
▷ Frame 5: 1003 bytes on wire (8024 bits), 1003 bytes captured (8024 bits)
▷ Ethernet II, Src: Cisco_31:07:33 (00:26:0b:31:07:33), Dst: Dell_c0:56:f0 (00:21:70:c0:56:f0)
▷ Internet Protocol Version 4, Src: 69.63.180.173, Dst: 172.16.0.122
▷ Transmission Control Protocol, Src Port: 443, Dst Port: 54595, Seq: 1, Ack: 170, Len: 937
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Handshake Protocol: Server Hello
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 74
    ◢ Handshake Protocol: Server Hello
        Handshake Type: Server Hello (2)
        Length: 70
        Version: TLS 1.0 (0x0301)
      ◢ Random: b9bb3b517aba70530291e8b0f97bb711647b94836658c94c...
          GMT Unix Time: Sep 28, 2068 19:56:17.000000000 대한민국 표준시
          Random Bytes: 7aba70530291e8b0f97bb711647b94836658c94c504630a2...
        Session ID Length: 32
        Session ID: 798e78f8199088e83fcf3e2ece32d14d26bc29eda5eb9149...
        Cipher Suite: TLS_RSA_WITH_RC4_128_MD5 (0x0004)
        Compression Method: null (0)
  ▷ TLSv1 Record Layer: Handshake Protocol: Certificate
  ▷ TLSv1 Record Layer: Handshake Protocol: Server Hello Done
```



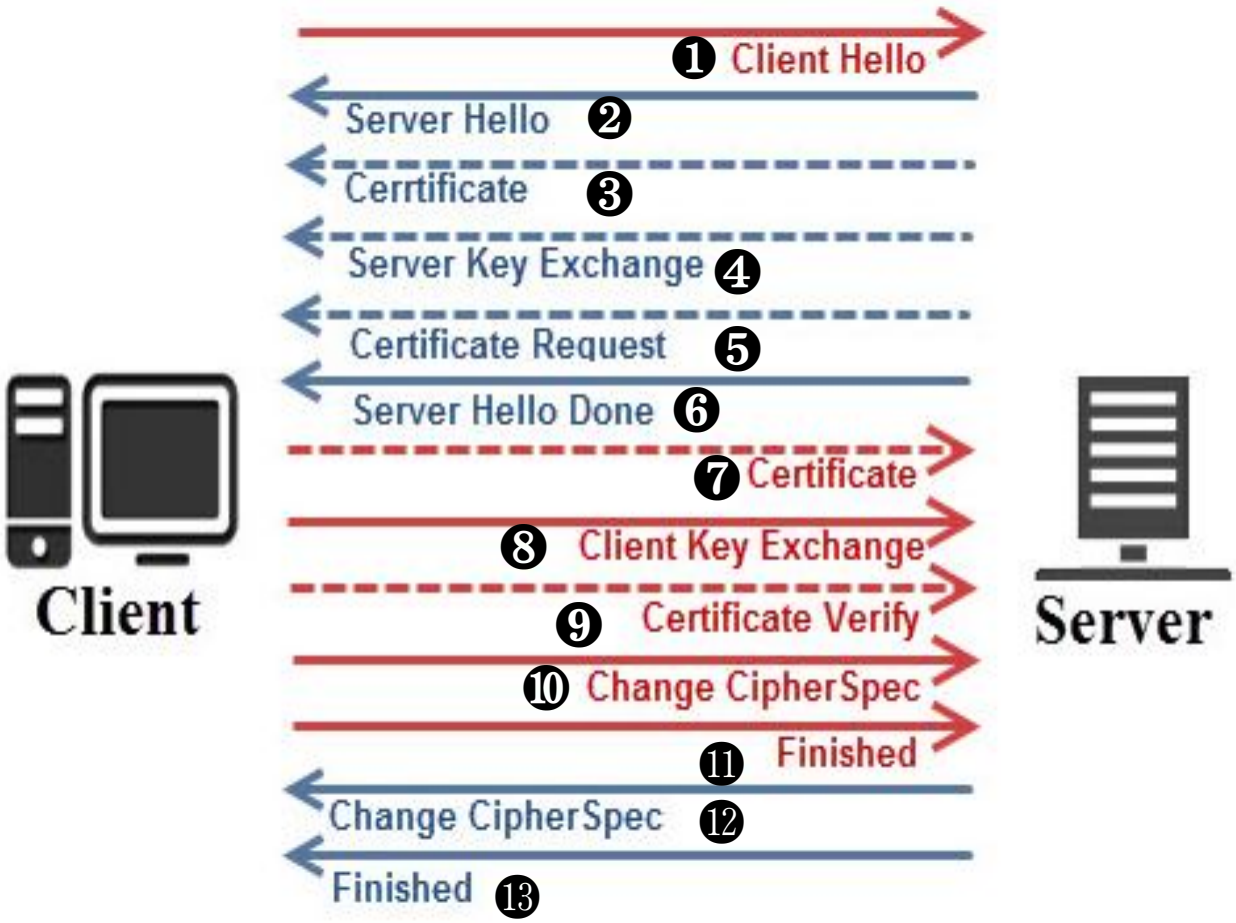

```
▷ Frame 5: 1003 bytes on wire (8024 bits), 1003 bytes captured (8024 bits)
▷ Ethernet II, Src: Cisco_31:07:33 (00:26:0b:31:07:33), Dst: Dell_c0:56:f0 (00:21:70:c0:56:f0)
▷ Internet Protocol Version 4, Src: 69.63.180.173, Dst: 172.16.0.122
▷ Transmission Control Protocol, Src Port: 443, Dst Port: 54595, Seq: 1, Ack: 170, Len: 937
◢ Secure Sockets Layer
  ▷ TLSv1 Record Layer: Handshake Protocol: Server Hello
  ▷ TLSv1 Record Layer: Handshake Protocol: Certificate
  ▷ TLSv1 Record Layer: Handshake Protocol: Server Hello Done
```

2 단계. Certificate message

```
▷ TLSv1 Record Layer: Handshake Protocol: Server Hello
◢ TLSv1 Record Layer: Handshake Protocol: Certificate
    Content Type: Handshake (22)
    Version: TLS 1.0 (0x0301)
    Length: 844
  ◢ Handshake Protocol: Certificate
      Handshake Type: Certificate (11)
      Length: 840
      Certificates Length: 837
    ◢ Certificates (837 bytes)
        Certificate Length: 834
      ◢ Certificate: 3082033e308202a7a00302010202030c183f300d06092a86... (id-at-commonName=login.facebook.com,id-at-organizationalU
        ◢ signedCertificate
            version: v3 (2)
            serialNumber: 792639
          ▷ signature (sha1WithRSAEncryption)
          ▷ issuer: rdnSequence (0)
          ▷ validity
          ▷ subject: rdnSequence (0)
          ◢ subjectPublicKeyInfo
            ◢ algorithm (rsaEncryption)
                Algorithm Id: 1.2.840.113549.1.1.1 (rsaEncryption)
            ▷ subjectPublicKey: 30818902818100d6114e489ea6f3a3611ac38d0b3e46a30c...
          ▷ extensions: 5 items
        ◢ algorithmIdentifier (sha1WithRSAEncryption)
            Algorithm Id: 1.2.840.113549.1.1.5 (sha1WithRSAEncryption)
          Padding: 0
          encrypted: 36104b5d5b134d9fc9b0b9a47d1131e528992f9a6bf2f74f...
```

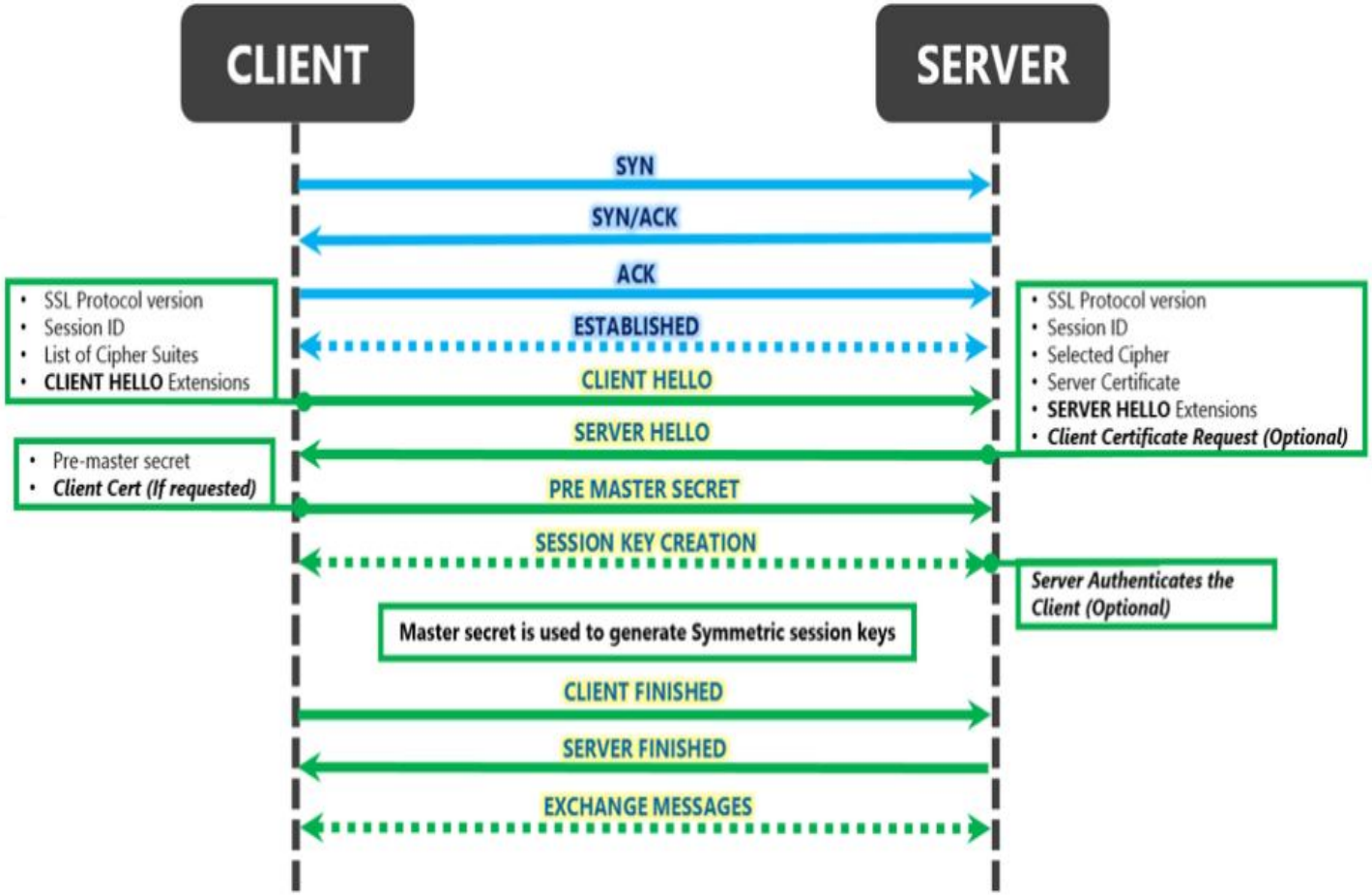
2 단계. Server Hello Done message

```
▷ Frame 5: 1003 bytes on wire (8024 bits), 1003 bytes captured (8024 bits)
▷ Ethernet II, Src: Cisco_31:07:33 (00:26:0b:31:07:33), Dst: Dell_c0:56:f0 (00:21:70:c0:56:f0)
▷ Internet Protocol Version 4, Src: 69.63.180.173, Dst: 172.16.0.122
▷ Transmission Control Protocol, Src Port: 443, Dst Port: 54595, Seq: 1, Ack: 170, Len: 937
◢ Secure Sockets Layer
  ▷ TLSv1 Record Layer: Handshake Protocol: Server Hello
  ▷ TLSv1 Record Layer: Handshake Protocol: Certificate
  ◢ TLSv1 Record Layer: Handshake Protocol: Server Hello Done
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 4
    ◢ Handshake Protocol: Server Hello Done
        Handshake Type: Server Hello Done (14)
        Length: 0
```



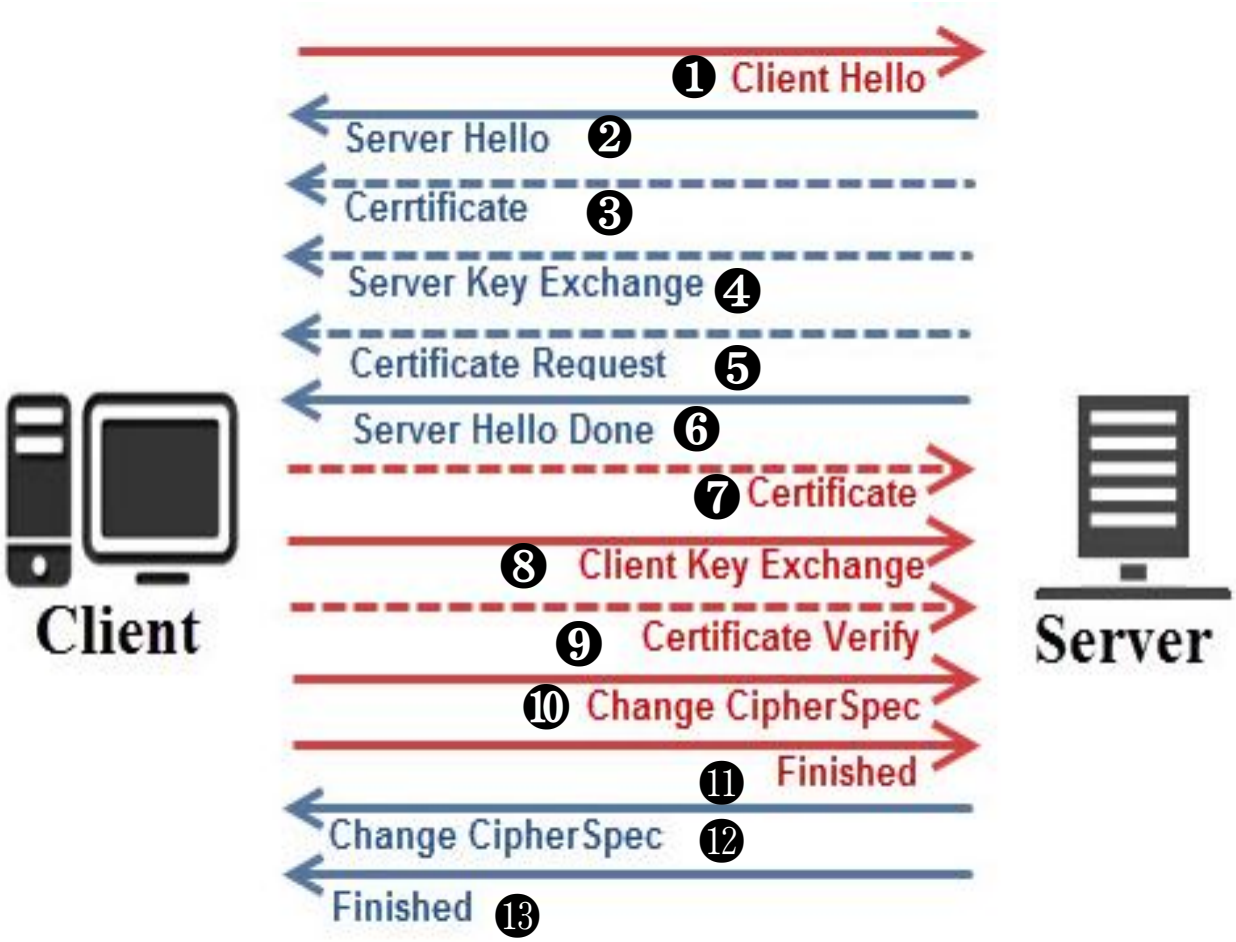
	Time	Source	Destination	Protocol	Total Length	Info
1	0.000000s	172.16.0.122	69.63.180.173	TCP	60	54595 → 443 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3019897…
2	0.089900s	69.63.180.173	172.16.0.122	TCP	64	443 → 54595 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 TSval=347…
3	0.000033s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=301989735 TSecr=347912…
4	0.000343s	172.16.0.122	69.63.180.173	TLSv1	221	Client Hello
5	0.089522s	69.63.180.173	172.16.0.122	TLSv1	989	Server Hello, Certificate, Server Hello Done
6	0.000031s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=170 Ack=938 Win=7744 Len=0 TSval=301989758 TSecr=34…
7	0.002848s	172.16.0.122	69.63.180.173	TLSv1	234	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
8	0.090444s	69.63.180.173	172.16.0.122	TLSv1	95	Change Cipher Spec, Encrypted Handshake Message
9	0.000533s	172.16.0.122	69.63.180.173	TLSv1	1034	Application Data
10	0.189619s	69.63.180.173	172.16.0.122	TCP	52	443 → 54595 [ACK] Seq=981 Ack=1334 Win=5473 Len=0 TSval=3479126142 TSecr=…
11	0.073201s	69.63.180.173	172.16.0.122	TLSv1	1233	Application Data
12	0.011497s	172.16.0.122	69.63.190.22	HTTP	679	GET /home.php? HTTP/1.1

3 단계.




```
▷ Frame 7: 248 bytes on wire (1984 bits), 248 bytes captured (1984 bits)
▷ Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: Cisco_31:07:33 (00:26:0b:31:07:33)
▷ Internet Protocol Version 4, Src: 172.16.0.122, Dst: 69.63.180.173
▷ Transmission Control Protocol, Src Port: 54595, Dst Port: 443, Seq: 170, Ack: 938, Len: 182
◢ Secure Sockets Layer
  ▷ TLSv1 Record Layer: Handshake Protocol: Client Key Exchange
  ▷ TLSv1 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec
  ▷ TLSv1 Record Layer: Handshake Protocol: Encrypted Handshake Message
```

```
▷ Frame 7: 248 bytes on wire (1984 bits), 248 bytes captured (1984 bits)
▷ Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: Cisco_31:07:33 (00:26:0b:31:07:33)
▷ Internet Protocol Version 4, Src: 172.16.0.122, Dst: 69.63.180.173
▷ Transmission Control Protocol, Src Port: 54595, Dst Port: 443, Seq: 170, Ack: 938, Len: 182
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Handshake Protocol: Client Key Exchange
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 134
    ◢ Handshake Protocol: Client Key Exchange
        Handshake Type: Client Key Exchange (16)
        Length: 130
      ◢ RSA Encrypted PreMaster Secret
          Encrypted PreMaster length: 128
          Encrypted PreMaster: 3b68c9a6fea0f7888813d309c8a1d81344b4b01f17d9a8ec...
  ◢ TLSv1 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec
      Content Type: Change Cipher Spec (20)
      Version: TLS 1.0 (0x0301)
      Length: 1
      Change Cipher Spec Message
  ◢ TLSv1 Record Layer: Handshake Protocol: Encrypted Handshake Message
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 32
      Handshake Protocol: Encrypted Handshake Message
```



No.	Time	Source	Destination	Protocol	Total Length	Info
1	0.000000s	172.16.0.122	69.63.180.173	TCP	60	54595 → 443 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=3019897…
2	0.089900s	69.63.180.173	172.16.0.122	TCP	64	443 → 54595 [SYN, ACK] Seq=0 Ack=1 Win=4140 Len=0 MSS=1380 WS=1 TSval=347…
3	0.000033s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=1 Ack=1 Win=5888 Len=0 TSval=301989735 TSecr=347912…
4	0.000343s	172.16.0.122	69.63.180.173	TLSv1	221	Client Hello
5	0.089522s	69.63.180.173	172.16.0.122	TLSv1	989	Server Hello, Certificate, Server Hello Done
6	0.000031s	172.16.0.122	69.63.180.173	TCP	52	54595 → 443 [ACK] Seq=170 Ack=938 Win=7744 Len=0 TSval=301989758 TSecr=34…
7	0.002848s	172.16.0.122	69.63.180.173	TLSv1	234	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
8	0.090444s	69.63.180.173	172.16.0.122	TLSv1	95	Change Cipher Spec, Encrypted Handshake Message
9	0.000533s	172.16.0.122	69.63.180.173	TLSv1	1034	Application Data
10	0.189619s	69.63.180.173	172.16.0.122	TCP	52	443 → 54595 [ACK] Seq=981 Ack=1334 Win=5473 Len=0 TSval=3479126142 TSecr=…
11	0.073201s	69.63.180.173	172.16.0.122	TLSv1	1233	Application Data
12	0.011497s	172.16.0.122	69.63.190.22	HTTP	679	GET /home.php? HTTP/1.1

```
▷ Frame 8: 109 bytes on wire (872 bits), 109 bytes captured (872 bits)
▷ Ethernet II, Src: Cisco_31:07:33 (00:26:0b:31:07:33), Dst: Dell_c0:56:f0 (00:21:70:c0:56:f0)
▷ Internet Protocol Version 4, Src: 69.63.180.173, Dst: 172.16.0.122
▷ Transmission Control Protocol, Src Port: 443, Dst Port: 54595, Seq: 938, Ack: 352, Len: 43
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Change Cipher Spec Protocol: Change Cipher Spec
      Content Type: Change Cipher Spec (20)
      Version: TLS 1.0 (0x0301)
      Length: 1
      Change Cipher Spec Message
  ◢ TLSv1 Record Layer: Handshake Protocol: Encrypted Handshake Message
      Content Type: Handshake (22)
      Version: TLS 1.0 (0x0301)
      Length: 32
      Handshake Protocol: Encrypted Handshake Message
```

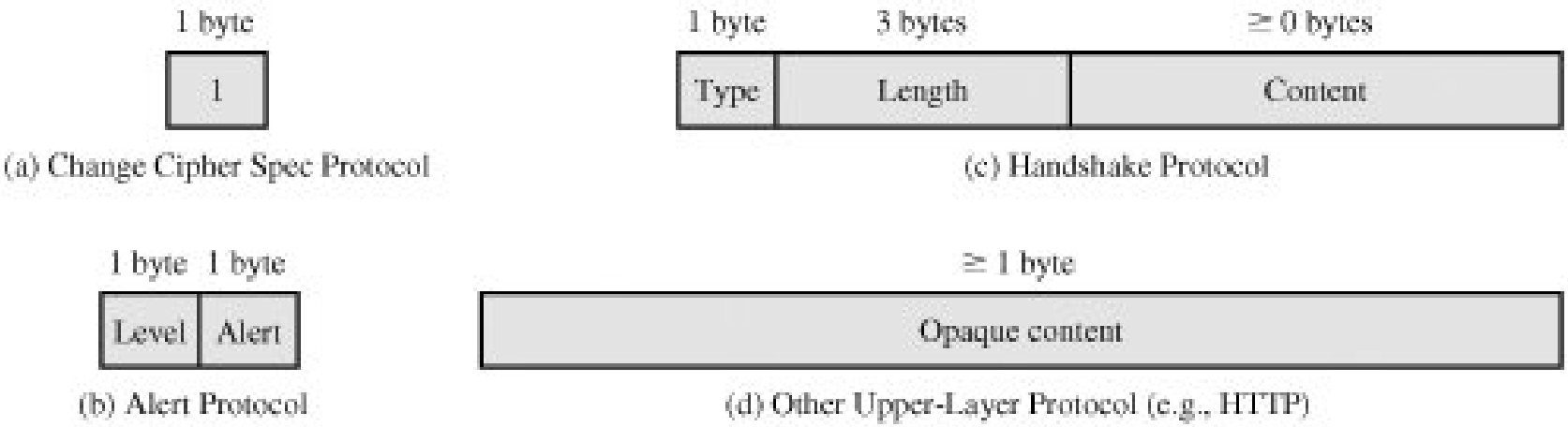
Key_block 생성

$$MS = MD5(S \mid SHA(\text{‘A’} \mid S \mid R_A \mid R_B)) \mid$$
$$MD5(S \mid SHA(\text{‘BB’} \mid S \mid R_A \mid R_B)) \mid$$
$$MD5(S \mid SHA(\text{‘CCC’} \mid S \mid R_A \mid R_B))$$

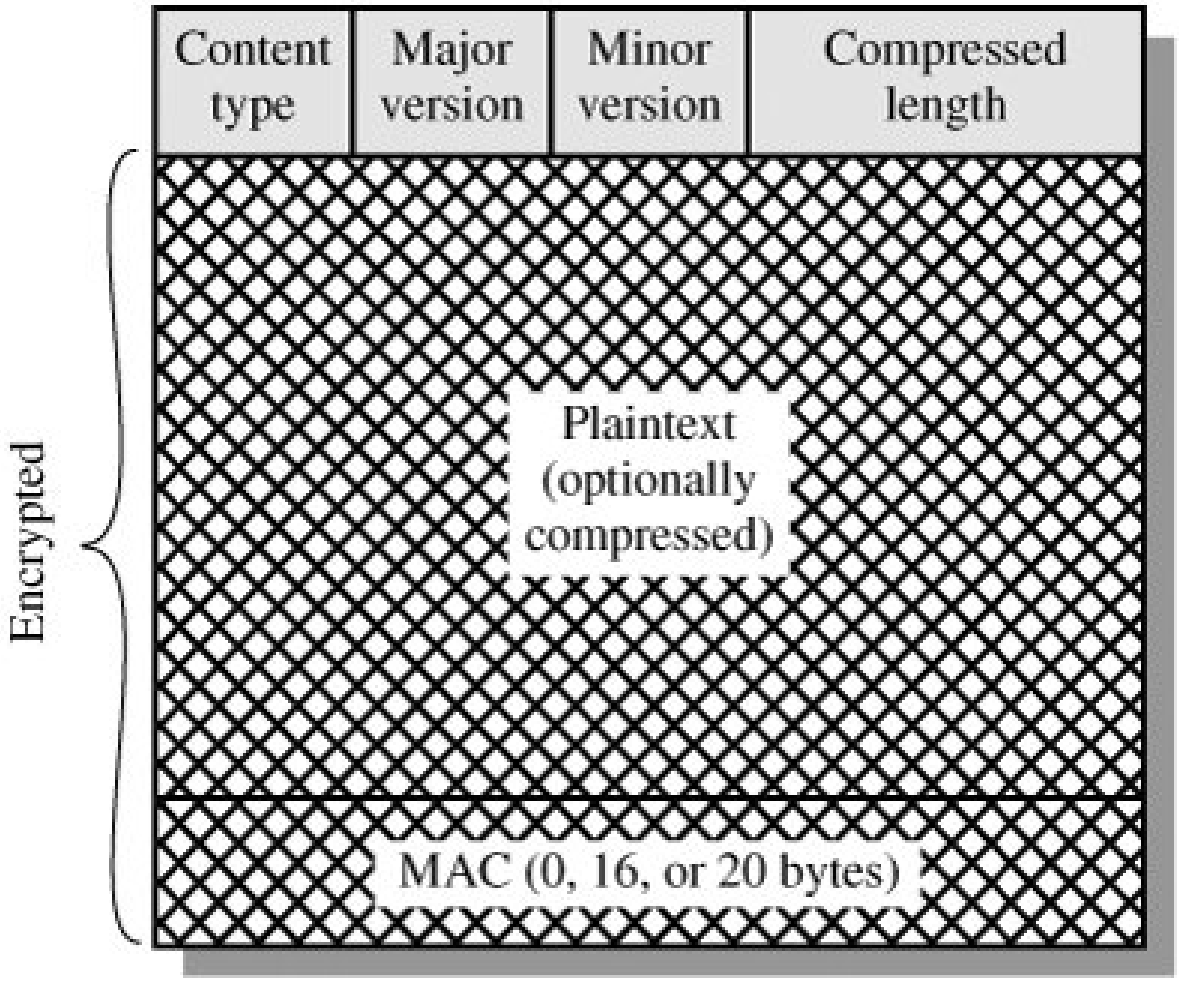
$$Key_block = MD5(MS \mid SHA(\text{‘A’} \mid MS \mid R_A \mid R_B)) \mid$$
$$MD5(MS \mid SHA(\text{‘BB’} \mid MS \mid R_A \mid R_B)) \mid$$
$$MD5(MS \mid SHA(\text{‘CCC’} \mid MS \mid R_A \mid R_B))$$

- Pre-master secret S
- 사용자가 보낸 난수 R_A
- 서버가 보낸 난수 R_B
- 문자열 ‘A’, ‘BB’, ‘ CCC’
- 해쉬 알고리즘 MD5, SHA

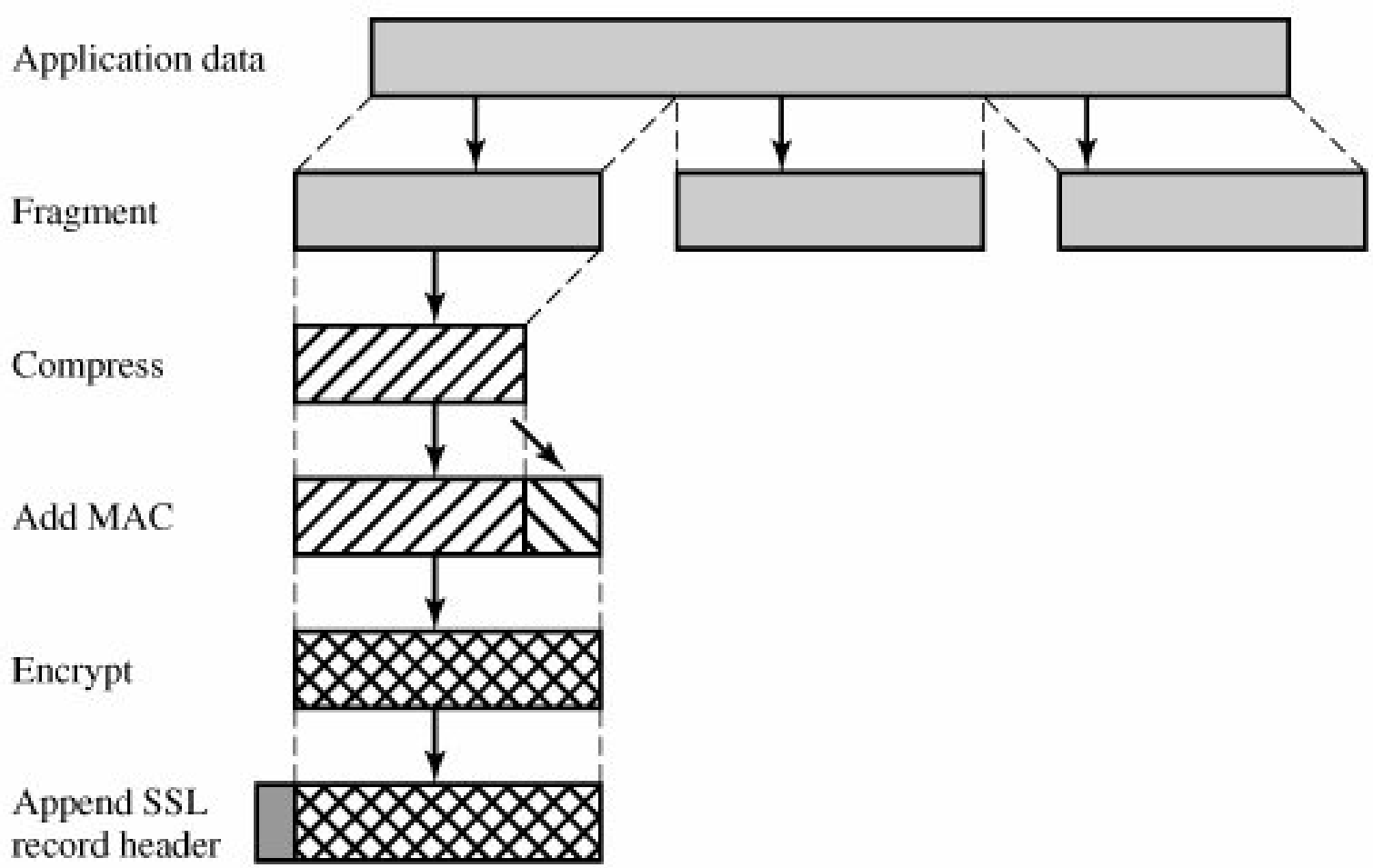
Record Protocol



SSL Record Format



SSL Record Protocol Operation



```
▷ Frame 9: 1048 bytes on wire (8384 bits), 1048 bytes captured (8384 bits)
▷ Ethernet II, Src: Dell_c0:56:f0 (00:21:70:c0:56:f0), Dst: Cisco_31:07:33 (00:26:0b:31:07:33)
▷ Internet Protocol Version 4, Src: 172.16.0.122, Dst: 69.63.180.173
▷ Transmission Control Protocol, Src Port: 54595, Dst Port: 443, Seq: 352, Ack: 981, Len: 982
◢ Secure Sockets Layer
  ◢ TLSv1 Record Layer: Application Data Protocol: http-over-tls
      Content Type: Application Data (23)
      Version: TLS 1.0 (0x0301)
      Length: 977
      Encrypted Application Data: 75926aa22236d0b22a72e9bdefc2d2579308f950d61ac3c7...
```

```
0030  00 79 aa 73 00 00 01 01  08 0a 11 ff ff 95 cf 5f   .y.s.... .......
0040  3f bf 17 03 01 03 d1 75  92 6a a2 22 36 d0 b2 2a   ?......u .j."6..*
0050  72 e9 bd ef c2 d2 57 93  08 f9 50 d6 1a c3 c7 4d   r.....W. ..P....M
0060  2c 11 0c 1a f1 2a a8 ae  83 ad f7 79 13 6e b9 46   ,....*.. ...y.n.F
0070  4d b4 f2 ce 70 87 60 9d  61 9f a0 43 a1 ee aa 64   M...p.`. a..C...d
0080  3b 4f 13 d0 87 89 cf 57  7c 6b 1a b8 5b 42 26 47   ;O.....W |k..[B&G
0090  e2 f4 4d 82 84 bd 41 f4  6a 2b 16 d6 0c aa 8e 84   ..M...A. j+......
00a0  e1 41 36 ee 12 8f 2b 98  48 0d 67 78 1c 2a 89 19   .A6...+. H.gx.*..
00b0  46 56 be e1 3a 02 bd 40  23 93 74 38 60 32 95 1e   FV..:..@ #.t8`2..
00c0  1f f8 60 cf ed 7f 8d 58  fa 25 c4 2e 33 3b 90 b7   ..`....X .%..3;..
00d0  6f 59 cc 8e ea a4 77 8d  f8 f1 f5 4e 82 fc 7e 20   oY....w. ...N..~ 
00e0  7c af d3 93 7b 80 4c d5  b8 be 5e d1 9a be 31 16   |...{.L. ..^...1.
00f0  4c 15 d4 ed 26 83 c2 31  0e f5 a9 03 1f e0 d9 28   L...&..1 .......(
0100  38 f3 a4 9c ac f1 d3 a6  2d c0 e8 fd be 6d 8d 43   8....... -....m.C
0110  33 59 73 62 98 87 0a 94  3f ee 15 93 82 ee 30 61   3Ysb.... ?.....0a
0120  96 fb 8a ef 1d 92 ba a0  32 cc 25 04 c7 f6 3f fd   ........ 2.%...?.
0130  a1 79 46 a8 84 85 e9 bc  67 c0 97 68 d4 9a 99 1a   .yF..... g..h....
0140  f2 16 9d 09 3c 00 93 1f  b9 a3 09 23 3f fc 6d cb   ....<... ...#?.m.
0150  c7 a1 cb 83 b4 23 43 ac  2b a8 71 7b b8 71 62 d5   .....#C. +.q{.qb.
0160  6e 93 fe ae de e3 94 f4  29 3f 0b 92 07 39 c8 0f   n....... )?...9..
```

Alert protocol

사용자나 서버가 프로토콜 진행 중에 문제를 감지하면 경고 메시지를 상대방에게 전송
치명적인(fatal) 경고와 비치명적인 경고 (fatal alert) 로 나뉨